

UNCONDITIONAL HYPERBOLICITY OF THE CRITICAL LAGUERRE FAMILY: AN ELEMENTARY AND FORMALLY VERIFIED PROOF

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ABSTRACT. For $d \geq 1$ let $C_d(w) = {}_1F_1(-d; \frac{1}{2}; w^2/4d)$ — up to normalization the Laguerre polynomial $L_d^{(-1/2)}$ at its critical scaling — and let

$$\Psi_d(w) = \sum_{k=0}^d (-1)^k \frac{d^{\underline{k}} M_k}{d^{\underline{k}} (2k)!} w^{2k}, \quad M_k = \frac{(2k)!}{4^k k!} e^{\gamma k - S_1(k)},$$

where $d^{\underline{k}}$ is the falling factorial, γ is Euler's constant and $S_1(k) = \sum_{j=1}^k H_{j-1}$ is the partial sum of harmonic numbers. We prove that for every $d \geq 1$ all $2d$ zeros of Ψ_d are real and simple. The numbers M_k are shown to be the moment sequence of an explicit compound-Poisson random variable $U \in (0, 1]$ with atom $\mathbb{P}(U = 1) = \sqrt{2/e}$, so that $\Psi_d(w) = \mathbb{E} C_d(\sqrt{U} w)$: hyperbolicity of the whole family is an exact averaging, not an asymptotic, phenomenon. The proof is elementary — real-variable throughout, using only the Kummer differential equation, an energy monotonicity along the real axis, and a sign-alternation count at the critical points of C_d — and it has been formally verified: the theorem is proved in Lean 4 against the mathlib library with zero `sorrys`, depending only on the three standard axioms. To our knowledge Ψ_d is, in this generality, a new explicit hyperbolic deformation of the Laguerre–Hermite family.

1. INTRODUCTION

A real polynomial is *hyperbolic* if all of its complex zeros are real. The hyperbolicity of the classical orthogonal families — Hermite, Laguerre, Jacobi — is classical [1, Thm. 3.3.1], and the linear operations preserving hyperbolicity were characterized by Pólya and Schur [2]. Deciding hyperbolicity of a *specific* explicitly given family that is not covered by such preserver theorems remains, in general, a genuinely hard problem: one must either locate all the zeros or manufacture enough sign changes.

This paper proves hyperbolicity, with simple zeros, for the following family. Throughout, $d^{\underline{k}} = d(d-1) \cdots (d-k+1)$ denotes the falling factorial, $H_n =$

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$\sum_{j=1}^n 1/j$ the harmonic numbers ($H_0 = 0$), γ Euler's constant, and

$$(1) \quad S_1(k) = \sum_{j=1}^k H_{j-1} = kH_{k-1} - (k-1) \quad (k \geq 1), \quad S_1(0) = 0.$$

Definition 1.1. For integers $k \geq 0$ and $d \geq 1$ set

$$(2) \quad M_k = \frac{(2k)!}{4^k k!} e^{\gamma k - S_1(k)},$$

$$(3) \quad C_d(w) = \sum_{k=0}^d (-1)^k \frac{d^k}{d^k (2k)!} w^{2k} = {}_1F_1\left(-d; \frac{1}{2}; \frac{w^2}{4d}\right),$$

$$\Psi_d(w) = \sum_{k=0}^d (-1)^k \frac{d^k M_k}{d^k (2k)!} w^{2k}.$$

By Kummer's transformation of the confluent hypergeometric polynomial, C_d is the Laguerre polynomial $L_d^{(-1/2)}$ at the argument scaling $w^2/4d$ that places its zeros at unit mean density — we call $\{C_d\}_{d \geq 1}$ the *critical Laguerre family*, and Ψ_d its *moment deformation* by the sequence (2). The deformation is of genuinely infinite order: the sequence (M_k) is strictly decreasing (the ratio test below reduces this to $\log(1+x) < x$), from $M_0 = 1$ through $M_1 = e^\gamma/2 = 0.89053\dots$ toward its limit $p := \sqrt{2/e} = 0.85776\dots$, so every coefficient of C_d is damped by a different factor. Nothing about (2) makes (M_k) visibly a multiplier sequence in the sense of Pólya–Schur, and we do not know a preserver-theoretic proof of Theorem 1.2.

Theorem 1.2 (Main theorem). *For every integer $d \geq 1$, all $2d$ zeros of Ψ_d are real and simple.*

The proof has two ingredients, both of independent interest.

First ingredient: the moments are a compound-Poisson law. The sequence (2) is exactly the moment sequence of an explicit random variable: let N be a Poisson point process on $(0, \infty)$ with the finite intensity measure

$$\lambda(dt) = \eta(t) \frac{e^{-t}}{1 - e^{-t}} dt, \quad \eta(t) = \frac{e^{t/2}}{t} - \frac{1}{1 - e^{-t}} > 0,$$

and set $U = e^{-\sum_j T_j}$, the product over the points of N . Then (Theorem 2.4)

$$\mathbb{E}[U^k] = M_k \quad (k \geq 0), \quad \mathbb{P}(U = 1) = p = \sqrt{2/e}.$$

Consequently, writing ν for the law of $v = \sqrt{U}$ and $\mu = \nu - p\delta_1$ for its non-atomic part,

$$(4) \quad \Psi_d(w) = \mathbb{E} C_d(\sqrt{U} w) = p C_d(w) + \int_{(0,1]} C_d(vw) d\mu(v),$$

with $\mu \geq 0$ and $\mu((0, 1]) = 1 - p$. The identity (4) is exact, coefficient by coefficient, at every finite d ; no limit is involved. Nor is the intensity λ

an ansatz: Section 2 opens by deriving it from the moment ratios of (2), through the classical integral representations of the digamma function.

Second ingredient: a one-line energy bound at the critical points. Writing $E = C_d^2 + C_d'^2$, the Kummer differential equation $C_d'' - (w/2d) C_d' + C_d = 0$ gives $E'(x) = (x/d) C_d'(x)^2 \geq 0$ on $x \geq 0$; since $E(0) = 1$, every critical point c of C_d satisfies $|C_d(c)| \geq 1$ and, crucially, the *whole segment* $[0, c]$ is dominated: $|C_d(vc)| \leq |C_d(c)|$ for $0 \leq v \leq 1$. Feeding this into (4) and using only $p > \frac{1}{2}$ shows that Ψ_d *copies the sign pattern* of C_d at the critical points of C_d . The classical interlacing data of the Laguerre zeros then forces d sign changes of Ψ_d on $(0, \infty)$, which (with evenness and the degree count) pins all $2d$ zeros to the real axis, each simple.

No complex analysis enters: no Rouché contour, no Hermite–Biehler theory, no saddle points. The argument runs entirely on the real line, which is also what made a complete formal verification practical.

Formal verification. Theorem 1.2 has been proved in Lean 4 [6] against the mathlib library [7]: the statement

```
theorem theorem_M (d : ℕ) (hd : 1 ≤ d) :
  ∀ z ∈ ((Psi d).map (algebraMap ℝ ℂ)).roots, z.im = 0
```

compiles with zero `sorrys` and depends only on the three standard axioms of the Lean/mathlib universe (`propext`, `Classical.choice`, `Quot.sound`). Section 5 describes the formalization; the complete source, with build instructions and the verification scripts, is available at

<https://github.com/michael-haufschild-gib/theorem-m>

Motivation. The identity (4) exhibits Ψ_d as an exact average of rescaled copies of C_d , and Theorem 1.2 says that this particular averaging preserves hyperbolicity at every degree. Read as a statement about the sequence (M_k) , it is a concrete instance of a natural question in the tradition of Pólya–Schur preserver theory: *which moment sequences of random variables in $[0, 1]$ act coefficientwise on the critical Laguerre family without destroying hyperbolicity?* The compound-Poisson mechanism of Section 2 — an atom of mass $p > \frac{1}{2}$ at the identity scale plus a positive residual — combined with the energy bound of Section 3 yields a sufficient criterion of striking simplicity, and the sequence (2), with its digamma–Binet moment structure and atom exactly $\sqrt{2/e}$, is a distinguished instance in which the criterion is endpoint-sharp (Remark 6.1). Beyond that, the theorem stands on its own: an explicit, nontrivially deformed Laguerre family that remains hyperbolic at every degree, with an elementary, machine-checked proof and an exact probabilistic mechanism behind it.

Organization. Section 2 constructs the compound-Poisson law and proves $\mathbb{E}[U^k] = M_k$. Section 3 collects the classical facts about C_d and proves the energy lemma. Section 4 proves Theorem 1.2. Section 5 documents the formal verification. Section 6 closes with remarks on sharpness and an open question.

2. THE COMPOUND-POISSON LAW BEHIND THE MOMENTS

Reverse-engineering the intensity. Before verifying anything, we explain where λ comes from: it is forced by the moments, and the harmonic numbers in (2) dictate its shape through the digamma function. The ratios of consecutive moments are elementary — by the telescoping $S_1(k) - S_1(k-1) = H_{k-1}$ of (1),

$$\frac{M_k}{M_{k-1}} = \frac{2k(2k-1)}{4k} e^{\gamma-H_{k-1}} = \left(k - \frac{1}{2}\right) e^{\gamma-H_{k-1}}.$$

Now look for a random variable $U = e^{-Y} \in (0, 1]$ with these moments, taking $Y \geq 0$ infinitely divisible — the simplest class with fully explicit Laplace transforms: $\log \mathbb{E}[e^{-kY}] = \int_0^\infty (e^{-kt} - 1) \lambda(dt)$ for an intensity measure λ to be found. Consecutive log-moment differences are then $\int (e^{-kt} - e^{-(k-1)t}) \lambda(dt)$, and the ansatz $\lambda(dt) = \eta(t) \frac{e^{-t}}{1-e^{-t}} dt$ makes the difference collapse to a single Laplace transform, since $(e^{-kt} - e^{-(k-1)t}) \frac{e^{-t}}{1-e^{-t}} = -e^{-kt}$. Matching the moment ratios above therefore amounts to solving

$$\int_0^\infty e^{-kt} \eta(t) dt = \psi(k) - \log\left(k - \frac{1}{2}\right), \quad k = 1, 2, \dots,$$

where the harmonic numbers have become the digamma function through $\psi(k) = H_{k-1} - \gamma$. This transform pair is classical: Gauss's integral represents ψ , Frullani's integral represents the logarithm (the half-integer shift $k - \frac{1}{2}$ producing the factor $e^{t/2}$), and subtracting the two integrands term by term reads off

$$\eta(t) = \frac{e^{t/2}}{t} - \frac{1}{1-e^{-t}}.$$

Everything to this point is bookkeeping. The one substantive fact — the fact that makes the random variable exist and the whole method work — is that this particular difference of classical integrands is *positive* and integrable against $\frac{e^{-t}}{1-e^{-t}} dt$. That is Lemma 2.1, and the rest of the section re-runs the derivation forwards, rigorously.

Lemma 2.1. *For $t > 0$ let $\eta(t) = \frac{e^{t/2}}{t} - \frac{1}{1-e^{-t}}$. Then*

$$\begin{aligned} \eta(t) &= \frac{2 \sinh(t/2) - t}{t(1-e^{-t})} > 0, & \eta(t) &= \frac{t}{24} + O(t^2) \quad (t \downarrow 0), \\ \eta(t) \frac{e^{-t}}{1-e^{-t}} &= O(t^{-1}e^{-t/2}) \quad (t \rightarrow \infty). \end{aligned}$$

In particular $\lambda(dt) := \eta(t) \frac{e^{-t}}{1-e^{-t}} dt$ is a finite positive measure on $(0, \infty)$.

Proof. Multiplying η by $t(1-e^{-t}) > 0$ gives $e^{t/2}(1-e^{-t}) - t = e^{t/2} - e^{-t/2} - t = 2 \sinh(t/2) - t > 0$ for $t > 0$, since $\sinh u > u$ for $u > 0$. The expansion at 0 is immediate from $2 \sinh(t/2) - t = t^3/24 + O(t^5)$ and $t(1-e^{-t}) = t^2 + O(t^3)$;

the decay at infinity from $2 \sinh(t/2) \leq e^{t/2}$. Both endpoints are integrable against $e^{-t}/(1-e^{-t}) dt \sim dt/t$ at 0 and $\sim e^{-t} dt$ at ∞ , whence λ is finite. $\square$

Definition 2.2. Let $N = \{T_j\}$ be a Poisson point process on $(0, \infty)$ with intensity λ (a.s. finitely many points, since λ is finite), and set

$$Y = \sum_j T_j \in [0, \infty), \quad U = e^{-Y} \in (0, 1].$$

The variable U is a compound-Poisson exponential functional; its atom at 1 is the no-point event, $\mathbb{P}(U = 1) = e^{-\lambda((0, \infty))} > 0$. By Campbell's theorem (the Laplace functional of a Poisson process, [5, Sec. 3.2]),

$$(5) \quad \mathbb{E}[U^k] = \mathbb{E}e^{-kY} = \exp\left(\int_0^\infty (e^{-kt} - 1)\lambda(dt)\right), \quad k \geq 0.$$

The transform of η is a Binet-type integral. We only need it at the positive integers, where it is elementary:

Lemma 2.3 (integer Binet identity). *For every integer $k \geq 1$,*

$$\int_0^\infty e^{-kt} \eta(t) dt = H_{k-1} - \gamma - \log\left(k - \frac{1}{2}\right).$$

Proof. Split $\eta(t)e^{-kt} = (e^{-(k-1/2)t} - e^{-t})/t + (e^{-t}/t - e^{-kt}/(1 - e^{-t}))$. Each part is absolutely integrable: the first is bounded near 0 (numerator = $O(t)$) and decays exponentially; the second is Gauss's integrand for the digamma function. Frullani's integral [4, Sec. 12.16] gives

$$\int_0^\infty \frac{e^{-(k-1/2)t} - e^{-t}}{t} dt = -\log\left(k - \frac{1}{2}\right),$$

and Gauss's formula [4, Sec. 12.3]

$$\int_0^\infty \left(\frac{e^{-t}}{t} - \frac{e^{-kt}}{1 - e^{-t}}\right) dt = \psi(k) = H_{k-1} - \gamma$$

for the digamma function $\psi = \Gamma'/\Gamma$ at the positive integer k . Adding the two evaluations gives the claim. $\square$

Theorem 2.4 (moment identification). $\mathbb{E}[U^k] = M_k$ for every integer $k \geq 0$, and $\mathbb{P}(U = 1) = p = \sqrt{2/e}$. Consequently, with ν the law of $v = \sqrt{U}$ and $\mu := \nu - p\delta_1$, μ is a positive measure concentrated on $(0, 1]$ with total mass $1 - p$, and for every $d \geq 1$ the exact decomposition (4) holds.

Proof. Ratios. For $k \geq 1$, by (5),

$$\log \frac{\mathbb{E}[U^k]}{\mathbb{E}[U^{k-1}]} = \int_0^\infty (e^{-kt} - e^{-(k-1)t}) \eta(t) \frac{e^{-t}}{1 - e^{-t}} dt = - \int_0^\infty e^{-kt} \eta(t) dt,$$

because $(e^{-kt} - e^{-(k-1)t})e^{-t}/(1 - e^{-t}) = -e^{-kt}$. By Lemma 2.3,

$$(6) \quad \frac{\mathbb{E}[U^k]}{\mathbb{E}[U^{k-1}]} = \left(k - \frac{1}{2}\right) e^{\gamma - H_{k-1}}.$$

On the other hand, from (2) and the telescoping $S_1(k) - S_1(k-1) = H_{k-1}$ of (1),

$$\frac{M_k}{M_{k-1}} = \frac{2k(2k-1)}{4k} e^{\gamma - H_{k-1}} = \left(k - \frac{1}{2}\right) e^{\gamma - H_{k-1}}.$$

Since $\mathbb{E}[U^0] = 1 = M_0$, induction gives $\mathbb{E}[U^k] = M_k$ for all k .

Atom. Since $U \in (0, 1]$, $U^k \downarrow \mathbf{1}_{\{U=1\}}$ pointwise, so by dominated convergence $\mathbb{P}(U = 1) = \lim_k \mathbb{E}[U^k] = \lim_k M_k$. Stirling's formula gives

$$\frac{(2k)!}{4^k k!} = \sqrt{2} k^k e^{-k} (1 + O(1/k)),$$

while $H_{k-1} = \log k + \gamma - \frac{1}{2k} + O(k^{-2})$ and (1) give $\gamma k - S_1(k) = -k \log k + k - \frac{1}{2} + O(1/k)$; multiplying, $M_k = \sqrt{2} e^{-1/2} + O(1/k) \rightarrow \sqrt{2/e}$.

Decomposition. $v = \sqrt{U} \in (0, 1]$ has an atom of mass exactly p at 1, so $\mu = \nu - p\delta_1 \geq 0$ is concentrated on $(0, 1]$ with mass $1 - p$ (in fact $\mu(\{1\}) = 0$, since the atom of ν at 1 has mass exactly p ; nothing below uses this, and we do not record it formally). Finally, for any $d \geq 1$, expanding the finite sum (3) and using $\mathbb{E}[(\sqrt{U})^{2k}] = \mathbb{E}[U^k] = M_k$,

$$\mathbb{E} C_d(\sqrt{U} w) = \sum_{k=0}^d (-1)^k \frac{d^k}{d^k (2k)!} \mathbb{E}[U^k] w^{2k} = \Psi_d(w),$$

which is (4) after splitting $\nu = p\delta_1 + \mu$. $\square$

Remark 2.5. Only three consequences of Theorem 2.4 are used below: $\mu \geq 0$, the mass $\mu((0, 1]) = 1 - p$, and the support $\text{supp } \mu \subseteq [0, 1]$. The specific shape of λ matters only through the closed form (2) of the moments. Note also $p^2 = 2/e > 1/4$, i.e.

$$(7) \quad p > \frac{1}{2},$$

the single inequality on which the entire sign-transfer argument runs (it amounts to $8 > e$).

3. THE CRITICAL LAGUERRE POLYNOMIAL AND ITS ENERGY

Lemma 3.1 (Hermite bridge and zeros). *Let He_n denote the probabilists' Hermite polynomials, $\text{He}_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}$. Then*

$$(8) \quad C_d(w) = \frac{\text{He}_{2d}(w/\sqrt{2d})}{\text{He}_{2d}(0)}, \quad \text{He}_{2d}(0) = (-1)^d (2d-1)!!.$$

Consequently C_d is an even polynomial of degree $2d$ with $C_d(0) = 1$, and all its zeros are real and simple:

$$-\rho_d < \cdots < -\rho_1 < 0 < \rho_1 < \cdots < \rho_d.$$

Proof. The quadratic transformation between Hermite and Laguerre polynomials [1, (5.6.1)], in confluent-hypergeometric form [3, Ch. 18], reads

$H_{2d}(x) = (-1)^d \frac{(2d)!}{d!} {}_1F_1(-d; \frac{1}{2}; x^2)$ for the physicists' polynomials H_n ; converting by $\text{He}_n(x) = 2^{-n/2} H_n(x/\sqrt{2})$ and substituting $x = w/\sqrt{4d}$ yields (8), the constant being $\text{He}_{2d}(0) = (-1)^d (2d)!/(2^d d!) = (-1)^d (2d-1)!!$. The zeros of Hermite polynomials are real and simple [1, Thm. 3.3.1], and 0 is not among them for even index since $\text{He}_{2d}(0) \neq 0$; evenness of He_{2d} arranges them symmetrically.

Remark on the formal proof. In the Lean formalization the identity (8) is not imported from the literature: both sides are even polynomials with constant term 1 whose even coefficient sequences satisfy the same first-order recurrence

$$(2k+1)(2k+2) a_{k+1} = (2k-2d) a_k / (2d)$$

forced by their respective differential equations, and the real-rootedness of He_n , with simplicity and the interlacing used below, is proved from scratch by induction on n through the recurrence $\text{He}_{n+1} = x\text{He}_n - \text{He}'_n$ (Section 5). $\square$

Lemma 3.2 (Kummer ODE). $C_d''(w) - \frac{w}{2d} C_d'(w) + C_d(w) = 0$.

Proof. With $u(z) = {}_1F_1(-d; 1/2; z)$ and $z = w^2/4d$, Kummer's equation $z u'' + (\frac{1}{2} - z) u' + d u = 0$ [3, Ch. 13] transforms under $C_d(w) = u(w^2/4d)$ by the chain rule: $C_d' = u' \cdot w/2d$ and $C_d'' = u'' w^2/4d^2 + u'/2d$, so, collecting terms (all derivatives of u evaluated at $z = w^2/4d$),

$$C_d'' - \frac{w}{2d} C_d' + C_d = \frac{u'' w^2}{4d^2} + \frac{u'}{2d} - \frac{u' w^2}{4d^2} + u = \frac{1}{d} \left(z u'' + \left(\frac{1}{2} - z\right) u' + d u \right) = 0. \quad \square$$

Lemma 3.3 (critical data). C_d' has exactly $2d-1$ zeros, all real and simple; on $[0, \infty)$ they are $0 = c_0 < c_1 < \dots < c_{d-1}$, interlacing with the zeros as

$$0 = c_0 < \rho_1 < c_1 < \rho_2 < \dots < c_{d-1} < \rho_d,$$

and the critical values alternate in sign:

$$(9) \quad \text{sgn } C_d(c_m) = (-1)^m, \quad m = 0, \dots, d-1.$$

For $d = 1$ the list of positive critical points is empty and (9) reduces to $C_1(0) = 1 > 0$.

Proof. C_d' has degree $2d-1$; Rolle's theorem places at least one of its zeros strictly between consecutive zeros of C_d , which accounts for $2d-1$ of them and therefore for all, each simple, one per gap. The gap $(-\rho_1, \rho_1)$ contains the critical point 0 (evenness). On (ρ_m, ρ_{m+1}) the polynomial C_d has no zero, hence constant sign; the sign on $(-\rho_1, \rho_1)$ is $+$ because $C_d(0) = 1$, and it flips across each simple zero ρ_m , so the sign on (ρ_m, ρ_{m+1}) is $(-1)^m$. Since $c_m \in (\rho_m, \rho_{m+1})$ for $1 \leq m \leq d-1$ and $c_0 = 0$, the display (9) follows. $\square$

Lemma 3.4 (energy monotonicity and segment domination). Let $E(x) = C_d(x)^2 + C_d'(x)^2$. Then $E'(x) = \frac{x}{d} C_d'(x)^2 \geq 0$ for $x \geq 0$, and $E(0) = 1$. Consequently, for every critical point $c \geq 0$ of C_d and every $v \in [0, 1]$,

$$(10) \quad |C_d(vc)| \leq |C_d(c)|, \quad \text{and} \quad |C_d(c)| \geq 1.$$

Proof. Differentiate and substitute the ODE of Lemma 3.2:

$$E' = 2C'_d (C_d + C''_d) = 2C'_d \cdot \frac{x}{2d} C'_d = \frac{x}{d} C_d'^2 \geq 0 \quad (x \geq 0).$$

$E(0) = C_d(0)^2 + C'_d(0)^2 = 1 + 0$. Monotonicity gives, for $0 \leq v \leq 1$,

$$C_d(vc)^2 \leq E(vc) \leq E(c) = C_d(c)^2,$$

the last equality because $C'_d(c) = 0$; and $C_d(c)^2 = E(c) \geq E(0) = 1$. $\square$

Remark 3.5. Lemma 3.4 is the real-axis case of an energy monotonicity that in fact holds along every horizontal line in $\mathbb{C}$ ($\partial_x(|C_d|^2 + |C'_d|^2) = (x/d)|C'_d|^2$, the imaginary part of w contributing no real part to the cross terms). Only the real-axis case is needed here, and only it was formalized. We also note that (10) with constant 1 is what makes the argument work with no room to spare; see Remark 6.1.

4. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.2. Fix $d \geq 1$ and abbreviate $p = \sqrt{2/e}$. Recall from (4) that $\Psi_d(x) = pC_d(x) + \int_{(0,1]} C_d(vx) \, d\mu(v)$ with $\mu \geq 0$ of total mass $1 - p$.

Step 1: Ψ_d copies the critical signs of C_d . Let $c \in \{c_0, \dots, c_{d-1}\}$ be a nonnegative critical point of C_d . By the segment domination (10),

$$\left| \Psi_d(c) - pC_d(c) \right| \leq \int_{(0,1]} |C_d(vc)| \, d\mu(v) \leq (1 - p) |C_d(c)| < p |C_d(c)|,$$

the last (strict) inequality by (7) and $|C_d(c)| \geq 1 > 0$. Hence $\Psi_d(c) \neq 0$ and

$$(11) \quad \text{sgn } \Psi_d(c_m) = \text{sgn } C_d(c_m) = (-1)^m, \quad m = 0, \dots, d-1.$$

Step 2: tail sign. The leading coefficient of Ψ_d is $(-1)^d d! M_d / (d^d (2d)!)$, of sign $(-1)^d$ since $M_d > 0$; the degree is exactly $2d$. Hence $\text{sgn } \Psi_d(x) = (-1)^d$ for all sufficiently large $x > 0$.

Step 3: count. By (11) and Step 2 the sequence of values

$$\Psi_d(c_0), \Psi_d(c_1), \dots, \Psi_d(c_{d-1}), \Psi_d(x_\infty) \quad (x_\infty \text{ large})$$

has signs $+, -, +, \dots, (-1)^{d-1}, (-1)^d$: it alternates through d strict sign changes. By the intermediate value theorem Ψ_d has at least d distinct zeros in $(c_0, \infty) = (0, \infty)$. Since Ψ_d is even, it has at least d further distinct zeros in $(-\infty, 0)$, and $\Psi_d(0) = M_0 = 1 \neq 0$. That is at least $2d$ distinct complex zeros of a polynomial of degree $2d$: there are exactly $2d$, all real, all simple. $\square$

Remark 4.1 ($d = 1$). The degenerate case is instructive: there are no positive critical points, the sign sequence is $\Psi_1(0) = 1 > 0$ followed by the tail sign $-$, and indeed $\Psi_1(w) = 1 - \frac{e^\gamma}{4} w^2$ has the two real simple zeros $\pm 2e^{-\gamma/2}$. The proof above covers this case verbatim — no step assumed $d \geq 2$.

5. THE FORMAL PROOF

Theorem 1.2 (in the realness form stated in Section 1) has been formalized in Lean 4 (v4.30.0) against mathlib (pinned at commit `c5ea0035` / tag `v4.30.0`).¹ The development comprises thirteen files, about 5,300 lines:

Defs	the polynomials Ψ_d , C_d , the sequences M_k , S_1
Structure	coefficient closed forms, degree, the Kummer ODE in polynomial form
Frullani	Frullani's integral for exponentials
Binet	the integer Binet identity (Lemma 2.3)
CPMeasure	the Lévy measure λ as a <code>withDensity</code> measure; finiteness
Moments, MomentsLimit	moment ratios; the Stirling limit $M_k \rightarrow p$
MuBridge	the finite-quadrature interface: residual moments $M_k - p$ from a measure
Capstone	discharge of the quadrature interface by the compound-Poisson pushforward
Energy	Lemma 3.4 (real energy monotonicity, segment domination)
Hermite	real-rootedness, simplicity and interlacing of He_n , by induction
CriticalData	Lemma 3.3 for C_d via the bridge (8); statement and proof of <code>theorem_M</code> and <code>theorem_M_aeval</code>
SignCount	the sign-alternation count, Step 3 above

Two design points deserve mention.

Measures, not samples. The formal development verifies every measure-level assertion of Section 2: positivity and finiteness of λ (a `withDensity` measure), the integer Binet identity as a genuine Lebesgue-integral statement, and the existence of the residual measure μ of (4) — obtained as a pushforward — with $\mu \geq 0$, concentration on $(0, 1]$, mass $1 - p$ and even moments $M_k - p$. What it does not do is represent U as the exponential functional $e^{-\sum_j T_j}$ of a Poisson point process. This is a deliberate boundary, not a gap: the proof of Theorem 1.2 consumes the law of U only through the listed measure data, so the point-process representation — the right way to *understand* the law, and the way it was found (Section 2) — contributes no additional logical content to the theorem. Formalizing it would enlarge the verified statement about the measure, not the verified theorem about Ψ_d .

Hermite theory from scratch. Rather than formalize the classical literature on Laguerre zeros, the development proves real-rootedness with simplicity and the full interlacing data for the probabilists' Hermite polynomials by induction on n via $\text{He}_{n+1} = x \text{He}_n - \text{He}'_n$: between consecutive simple roots of He_n the derivative term forces a sign change of He_{n+1} , and two tail roots are produced by the parity of the leading terms; a counting argument then pins exactly $n + 1$ simple roots. The identity (8) transports this package to C_d by a first-order coefficient recurrence, avoiding any appeal to unformalized special function theory.

¹Source, build instructions and verification scripts: <https://github.com/michael-haufschild-gib/theorem-m>.

The final check reports, for the headline theorem and its evaluation-form companion alike,

```
'TheoremM.theorem_M' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'TheoremM.theorem_M_aeval' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

with zero `sorry`s in the tree (8489 compiled jobs). These three axioms are the standard foundation of all of `mathlib`; in particular no numerical computation, floating point or interval arithmetic enters the formal proof anywhere — all constants, including $p = \sqrt{2}/e$ and the inequality (7), are handled exactly.

The formalization was produced jointly by two AI research agents (Claude “Fable” and GPT) working under the author’s direction in a mutual-audit protocol (disjoint file ownership, cross line-audits of each other’s proofs), with the Lean kernel as the final referee. In addition, the development has been re-verified independently of Lean’s own kernel: the proof terms were exported with the upstream `lean4export` tool (v4.30.0) and re-checked by Nanoda (`nanoda_lib`, commit `f58f2f6d`), an independent implementation of the Lean 4 kernel in Rust,² which reports “*Checked 48241 declarations with no errors*” for an export containing `theorem_M` and `theorem_M_aeval`, with the permitted axioms restricted to exactly the three listed above.

6. CONCLUDING REMARKS

Remark 6.1 (sharpness). The budget of Step 1 is sharp at the endpoint $v \rightarrow 1^-$: numerically, for every d tested ($1 \leq d \leq 300$), the worst ratio $|\Psi_d(c_m) - pC_d(c_m)| / p|C_d(c_m)|$ over the critical points equals $(1 - p)/p = 0.16582\dots$ to all computed digits — the measure μ concentrates its action at v near 1, where the domination (10) costs nothing. The proof therefore has a uniform factor-6 margin, but no asymptotic slack is hiding anywhere: the mechanism is the strict inequality $p > \frac{1}{2}$, used once.

Remark 6.2 (simplicity vs. the formal statement). The paper proof gives realness *and simplicity* (Step 3 produces $2d$ *distinct* zeros). The formalized statement asserts realness of every complex root; distinctness is present in the formal development’s counting layer but was not exported into the headline statement.

Remark 6.3 (a question). The moment sequence (2) acts coefficientwise on C_d and preserves hyperbolicity for every d . Is $(M_k)_{k \geq 0}$ a Pólya–Schur multiplier sequence (i.e. does it preserve hyperbolicity of *all* hyperbolic even polynomials)? The compound-Poisson representation proves much less — it is tied to the energy structure of C_d — and we do not know the answer; a negative answer would make Theorem 1.2 more interesting, not less.

²<https://github.com/leanprover/lean4export> and https://github.com/ammkrn/nanoda_lib.

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